

sections:

- block: markdown

content:

title: Research Areas

subtitle: "

text: |

Our research focuses on advancing computing systems by enhancing **performance, availability, reliability, cost-efficiency, power consumption, and runtime operations**. Our approach combines mathematical modelling with large-scale cloud experimentation, and we frequently develop proofs-of-concept and prototypes.

design:

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- block: markdown

content:

title:

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{{< spoiler text="🖥️ Serverless Computing" >}}

Performance modelling, simulation, and optimization of serverless (FaaS) platforms — including cold-start analysis, autoscaling, function placement, and cost/performance trade-offs.

{{< /spoiler >}}

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{{< spoiler text="🧠 Machine Learning Systems" >}}
Predicting deep neural network training time, convergence prediction
across architectures, SLA-aware inference serving, and ML-based
autoscaling for cloud applications.
{{< /spoiler >}}

{{< spoiler text="🌐 Microservice Platforms" >}}
Performance modelling of microservice platforms, software diversity and
multi-versioning to engineer reliable and performant systems, and
efficient provisioning of microservices.
{{< /spoiler >}}

{{< spoiler text="🔗 Cyber-Physical Systems" >}}
Self-managing IoT platforms, end-to-end IoT application management,
intrusion detection for smart grids, and big-data analytics for smart
transportation.
{{< /spoiler >}}

{{< spoiler text="☁️ Cloud Software Systems" >}}
Adaptive and self-managing cloud applications, elastic containerized
applications for DevOps, autoscaling and monitoring as a service, and
analytical performance/availability models of cloud data centers.
{{< /spoiler >}}

{{< spoiler text="⚙️ Operating Systems" >}}
Runtime operations, resource management, and controller-based
optimization (e.g., adaptive PID controllers) for containerized
software systems.
{{< /spoiler >}}

{{< spoiler text="🔗 Distributed Ledgers" >}}
Performance evaluation of blockchain systems, DAG-based distributed
ledgers for smart communities, blockchain-based serverless platforms,
and peer-to-peer energy trading frameworks.
{{< /spoiler >}}

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design:

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- block: collection

content:

title: Related Publications

text: "

count: 6

filters:

folders:

- publication

order: desc

design:

view: citation

columns: '1'
